

Lecture #14

Content to be covered

- ❑ Optical activity
 - ❖ Biot's law for optical rotation
 - ❖ Fresnel's Theory of Optical Rotation
 - ❖ Optically active substances
 - ❖ Specific Rotation

Resources to be consulted

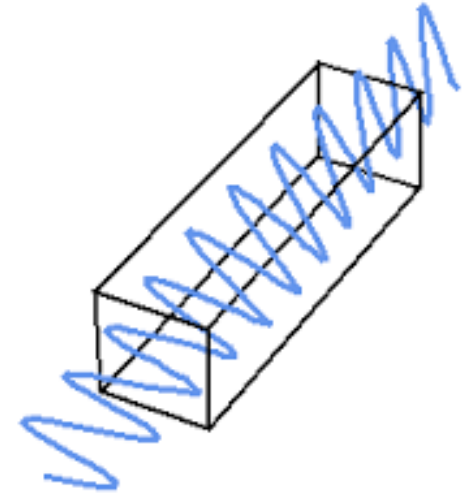
- ❑ OPTICS by Ajay Ghatak

Optical Activity

When a beam of plane polarized light propagates through certain substances or crystals, the plane of polarization of the emergent beam rotated through a certain angle. This phenomenon of rotation of the plane of vibration is called **rotatory** polarization and this property of the crystal (substance) is called **optical activity** or optical rotation and substances which show this property are called **optically active substances**.

There are two type of optically active substances:

- ❖ **Right handed or dextrorotatory:** Rotate the plane of vibration in the clockwise direction as seen by the observer facing the source. Examples: Sodium chlorate, cane sugar.
- ❖ **Left handed or laevorotatory:** Rotate the plane of vibration in the anti-clockwise direction as seen by an observer facing the source. Examples: Fruit sugar, turpentine.



Angle of rotation depends on the length and concentration of the medium.

Other Examples: Quartz crystals (Some may be dextro- and some laevo-rotatory. Calcite is optically inactive).

Biot's law for optical rotation

- * $\theta \propto \ell$ θ : angle of rotation of the plane of vibration for any given wavelength.
 ℓ : length of the optically active medium traversed.
- * In case of solution or vapours
 $\theta \propto C$, C : concentration of the solution or vapour
- * The angle of rotation is inversely proportional to the square of the wavelength for a given length of the optically active substance.
- * The total rotation produced by a number of optically active substances is equal to the algebraic sum of the individual rotations.

$$\theta = \theta_1 + \theta_2 + \theta_3 + \dots = \sum_i \theta_i$$

The anticlockwise rotations are taken +ve ;
while the clockwise rotations are taken -ve.

Applications:

1. To find the percentage of optically active material present in the solution.
2. The amount of sugar present in blood of a diabetic patient determined by measuring the angle of rotation of the plane of polarization.

Fresnel's Theory of Optical Rotation

Fresnel's theory of optical rotation by an optically active substance is based on the fact that any **plane polarized light may be considered as resultant of two circularly polarized vibrations rotating in opposite direction with the same velocity or frequency.**

$$\vec{E}_R = \hat{i}E_0 \cos(kz - \omega t) + \hat{j}E_0 \sin(kz - \omega t)$$

$$\vec{E}_L = \hat{i}E_0 \cos(kz - \omega t) - \hat{j}E_0 \sin(kz - \omega t)$$

$$\text{But } \vec{E} = \vec{E}_R + \vec{E}_L \Rightarrow \vec{E} = 2\hat{i}E_0 \cos(kz - \omega t) \quad \text{Linearly Polarized Light}$$

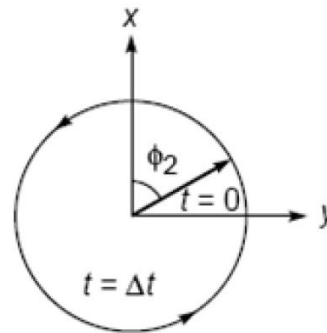
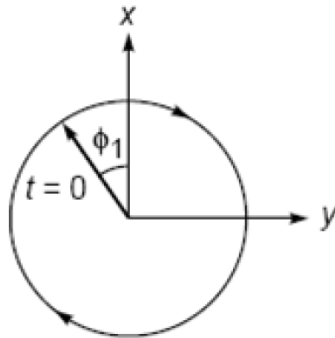
This explanation was based on the following **assumptions**:

A plane polarized light falling on an optically active medium along its optic axis splits up into two circularly polarized vibrations of equal amplitudes and rotating in opposite directions—one clockwise and other anticlockwise.



Fresnel's Theory of Optical Rotation

- ❑ In an optically inactive substance, these components travel with the same speed. Hence, at emergence, produce a plane polarized light without any rotation of the plane of polarization.
- ❑ In an optically active crystal like quartz, two circular components travel with different speeds so that relative phase difference is developed between them on transmission through the crystal.



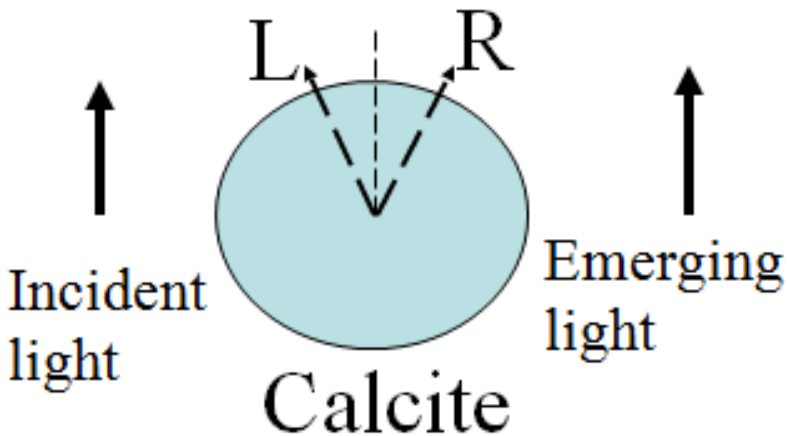
- ❖ In **dextrorotatory substance** $v_R > v_L$ and in **laevorotatory substance** $v_L > v_R$.
- ❖ On emergence from an optically active substance the two circular vibrations recombine to give plane polarized light whose plane of vibration has been rotated w.r.t that of incident light through a certain angle depends on the phase diff between the two vibrations.

Fresnel's Theory of Optical Rotation

Optic axes is perpendicular to refracting face.

$$\text{PPL} = \text{LCP} + \text{RCP}$$

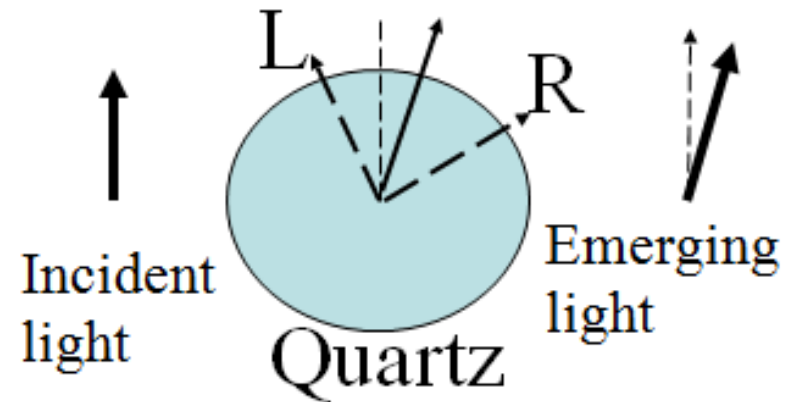
- Optically inactive material



Same angular velocity of R and L. it means equal angles on both sides.

Net rotation is zero

- Optically active material



Angular velocity of R is more than that of L. therefore emergent light will have a net rotation of emergent light by an amount $\theta = \delta/2$.

Mathematical Treatment

RCP propagating in z-direction: $E_{Rx} = E_0 \cos(k_R z - \omega t)$ and $E_{Ry} = E_0 \sin(k_R z - \omega t)$

LCP propagating in z-direction: $E_{Lx} = E_0 \cos(k_L z - \omega t)$ and $E_{Ly} = -E_0 \sin(k_L z - \omega t)$

where $k_R = \frac{\omega}{c} \mu_R$ and $k_L = \frac{\omega}{c} \mu_L$

Now $E_x = E_{Rx} + E_{Lx} = E_0 \cos(k_R z - \omega t) + E_0 \cos(k_L z - \omega t)$

$$\Rightarrow E_x = 2E_0 \cos\left[\frac{k_R z + k_L z}{2} - \omega t\right] \cos\left(\frac{k_R z - k_L z}{2}\right)$$

Similarly $E_y = 2E_0 \cos\left[\frac{k_R z + k_L z}{2} - \omega t\right] \sin\left(\frac{k_R z - k_L z}{2}\right)$

Now take $\frac{E_y}{E_x} = \tan \theta = \tan\left[\frac{k_R - k_L}{2} t\right] \Rightarrow \theta = \frac{k_R - k_L}{2} t = \frac{\omega}{c} \frac{\mu_R - \mu_L}{2} t$

or $\theta = \frac{\pi}{\lambda} (\mu_R - \mu_L) t$ where, t = thickness of optically active substance

For optically active substances

μ_R : the refractive index of clockwise vibration

μ_L : the refractive index of anti-clockwise vibration

t : the thickness of the quartz plate,

Thus, the path difference between the two components is given by $\Delta = (\mu_L \sim \mu_R)d$

Corresponding phase difference will be $\delta = \frac{2\pi}{\lambda}(\mu_L \sim \mu_R)d$

Angle of rotation of plane of vibration will be

$$\theta = \frac{\delta}{2} = \frac{\pi}{\lambda}(\mu_L \sim \mu_R)d = \frac{\pi}{\lambda} \left(\frac{c}{v_L} \sim \frac{c}{v_R} \right) d$$

In case of left handed optically substances $v_L > v_R$

In case of right handed optically substances $v_L < v_R$

Specific Rotation

The rotation (in degrees) produced by a path of one decimeter length of a substance of unit density for a given wavelength at a given temperature is defined as specific rotation.

If θ is the rotation produced by l decimeter length of an optically active substance having concentration of its solution c gm/cm³ then specific rotation at a given temperature T for a given wavelength λ is expressed as:

$$S(T, \lambda) = \frac{\theta}{lc}$$

(If l is in cm)

$$S(T, \lambda) = \frac{10\theta}{lc}$$

The unit of specific rotation is $\text{deg.}(\text{decimeter})^{-1}(\text{gm/cc})^{-1}$

The molecular rotation is given by the specific rotation and molecular weight.